

INTERNATIONAL CYBERSECURITY AND
DIGITAL FORENSICS ACADEMY

Week 4 Practical Laboratory

Linux Security Monitoring and Auditing Report

From Technical Evidence to Governance Assurance

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Course	GRC102 – Information Security Governance
Module	Module 4 – Monitoring and Auditing Security Controls
Environment	ICDFA Kali Linux practice VM (hostname: icdfa), authorised assigned system
Evidence collection period	01 October 2026 23:37 WAT to 02 October 2026 03:37 WAT
Submission Due	02 October 2026, 11:59 PM WAT

Submission mode: Written report + supporting evidence

Platform: ICDFA Ubuntu Practice Hub / authorised assigned Linux VM

1. Executive Summary

This report presents the results of a security monitoring and audit review of the assigned Linux system, carried out using auditd, journalctl log analysis and a Lynis 3.1.4 system audit. The evidence was mapped to control objectives, owners, thresholds and remediation actions.

Overall control posture: Partially effective. Detective controls are working: auditd is active, five custom rules are loaded, and user activity is attributable even after privilege escalation. Preventive and response controls are weaker. Lynis reported a hardening index of 65/100 with one warning, and several monitoring gaps were identified.

Top three findings

- W4-F02 (High): No active host firewall rules. Lynis warning FIRE-4512 reports iptables modules loaded but no rules active.
- W4-F01 (Moderate): /var/log/auth.log and /var/log/syslog do not exist. The auth_failures audit rule watches a file that is absent, so it provides no coverage and could give false assurance.
- W4-F03 / W4-F04 (Moderate): Password policy controls are not configured (Lynis AUTH-9286, AUTH-9262, AUTH-9282, AUTH-9230) and there is no automated response to repeated authentication failures (DEB-0880). Three historical authentication failures were logged but not acted on.

No evidence of compromise or unauthorised activity was observed. All privileged activity found in the logs on 02 October 2026 matched the analyst's own lab commands.

2. Scope and Authorisation

All activity was performed only inside the Linux virtual machine assigned for training (ICDFA Kali Linux practice VM, hostname icdfa, user icdfa). No external systems, production infrastructure or other devices were targeted.

- In scope: auditd configuration and audit events, systemd journal and equivalent log sources, Lynis security assessment of the assigned VM, and governance mapping of the evidence.
- Out of scope: SIEM or SOAR deployment (Module 4 is conceptual), penetration testing, and any system other than the assigned VM.
- Constraints observed: no changes were saved to /etc/passwd or /etc/shadow, logging was not disabled, and no evidence was deleted.

Limitation: the VM shut down unexpectedly during the first Lynis run (caused by a pending host update). Lynis was re-run to completion; the results reported here come from the completed run that finished at 2026-10-02 03:25:35 WAT.

3. Methodology

3.1 auditd

Service status was checked with systemctl. A dedicated rules file (/etc/audit/rules.d/custom.rules) was created containing watches on /etc/passwd and /etc/shadow, execve monitoring for 64-bit and 32-bit programs, and a watch on /var/log/auth.log. The rules were loaded by restarting auditd and verified with auditctl -l. Benign events were generated by opening /etc/passwd in an editor without saving and running ls /tmp. Evidence was retrieved with ausearch (by key) and aureport.

3.2 Log analysis

The systemd journal was reviewed with journalctl (today's entries, error priority, warning priority, _COMM=sudo). Because /var/log/auth.log and /var/log/syslog do not exist on this system, journalctl and grep against the journal were used as the documented equivalent sources.

3.3 Lynis

Lynis 3.1.4 was run with sudo lynis audit system. The hardening index, warnings and suggestions were extracted from /var/log/lynis-report.dat.

3.4 Governance mapping

Findings were mapped to a control objective, owner, observed status, KPI/KRI threshold, risk significance, remediation action and retest requirement, then to SIEM and continuous-monitoring concepts.

4. Findings and Evidence

4.1 Evidence Bundle 1: auditd configuration and events

ITEM	OBSERVATION
Service status	auditd active (running) since Thu 2026-10-01 23:37:54 WAT, auditd 4.1.2. Startup log noted: "No plugins found, not dispatching events".
Loaded rules (auditctl -l)	-w /etc/passwd -p rwx -k passwd_changes -w /etc/shadow -p rwx -k shadow_changes -a always,exit -F arch=b64 -S execve -F key=program_execution -a always,exit -F arch=b32 -S execve -F key=program_execution -w /var/log/auth.log -p wa -k auth_failures (auditctl displays -k as -F key= for the execve rules; they are equivalent.)
aureport summary (23:37:54 to 00:18:16 WAT)	14,167 events; 102 commands; 24 configuration changes; 0 account/group changes; 0 logins; 0 failed logins; 1 authentication; 0 failed authentications; 27 failed syscalls.

ITEM	OBSERVATION
aureport -k --summary	program_execution: 24,507; passwd_changes: 617; shadow_changes: 4.
ausearch -k passwd_changes	02/10/2026 00:29:41 WAT: sudo opened /etc/passwd (openat, O_RDONLY, success) with auid=icdfa, euid=root, tty=pts2, ses=3.
ausearch -k shadow_changes	00:29:43 and 00:32:12 WAT: /usr/sbin/unix_chkpwd (icdfa chkexpiry) opened /etc/shadow read-only, auid=icdfa.
ausearch -k program_execution	Captured execve events for cut, head, ip and grep (auid=icdfa, tty=none), indicating background shell or desktop helper activity.
ausearch -k auth_failures	Not meaningful: /var/log/auth.log does not exist on this system (see W4-F01).

Interpretation

The audit record shows that user icdfa (auid=icdfa, session 3, terminal pts2) used sudo to access sensitive account files. At 00:29:41 WAT on 02/10/2026, sudo opened /etc/passwd read-only, and at 00:29:43 WAT the password-check helper (unix_chkpwd) opened /etc/shadow read-only. Both actions succeeded and no write or modification events were recorded, which is consistent with the test in which /etc/passwd was opened and closed without saving. The record keeps the original user's identity (auid=icdfa) even though the process ran with root privileges (euid=root).

This matters to security governance because it shows that access to credential files and privileged activity are logged and traceable to a named individual, supporting accountability, access-control and change-management assurance, and allowing management to confirm that no unauthorised changes were made to user accounts. It also shows a tuning issue: read-level watches generate high volume (617 passwd_changes events) and should be filtered before forwarding to a SIEM to avoid alert fatigue.

4.2 Evidence Bundle 2: Linux log analysis

Review performed 02 October 2026 (journal reviewed at 00:56 WAT, authentication checks at 01:15 WAT, warnings at 01:22 WAT). Equivalent sources: /var/log/auth.log and /var/log/syslog are absent, so the systemd journal was used.

Event / Condition	Evidence Source	Time (WAT)	User / Service	Security Significance	Recommended Action
Privileged commands via sudo	journalctl _COMM=sudo	02 Oct 00:56:02, 00:58:51	icdfa to root	Privileged use is logged and attributable	Review sudo logs periodically; alert on unexpected users or commands
Authentication failure (sudo)	journalctl grep	10 May 21:20:11	icdfa (pts/0)	Isolated failure, consistent with a mistyped password	Alert at threshold (e.g. 5 in 10 min)
Authentication failure (polkit)	journalctl grep	20 Sep 02:47:01	icdfa (polkit)	As above; logged but no response	As above
Authentication failure (lightdm)	journalctl grep	24 Sep 20:09:02	icdfa (lightdm)	As above; failed desktop login	As above
Login-screen session release error	journalctl -p err	19 Sep 04:13:58	lightdm / pam_systemd	Minor session fault; no sign of attack	Monitor for recurrence
Missing GNOME evolution service	journalctl -p err	19 Sep 04:14:19	obexd	Unused desktop component, low risk	Remove unneeded services to reduce attack surface
Audit log rotation	journalctl --since today	02 Oct 00:41:29	auditd	Evidence is rotated; retention affects availability	Confirm retention meets policy; archive logs
Deprecated SysV script for inetsim	journalctl -p warning	01 Oct 23:22 to 23:37	systemd-sysv-generator	Legacy unmaintained service increases configuration risk	Replace with a native systemd unit or remove if unused
Realtime scheduling thread starved	journalctl -p warning	01 Oct 23:36:09	rtdk-daemon	Operational, not a direct security risk	Monitor; investigate if recurring
IMA duplicate measurements not logged	journalctl -p warning	02 Oct 00:22:41	kernel	Minor gap in integrity-measurement logging	Review IMA configuration if integrity monitoring is required
No auth.log / syslog files	ls, less output	02 Oct 01:10 / 01:22	System configuration	File-based log tooling would see nothing	Forward journal to SIEM or enable file-based logging

A search for "failed password" returned no genuine events (the only match was the analyst's own search command). A search for "authentication failure" returned three historical events for user icdfa. The isolated pattern over about five months suggests mistyped credentials rather than a brute-force attempt, but confirms that failed authentication is logged and attributable.

4.3 Evidence Bundle 3: Lynis security assessment

Item	Result
Lynis version	3.1.4
Audit completed	2026-10-02 03:25:35 WAT (sudo lynis audit system)
Hardening index	65 / 100
Warnings	1: FIRE-4512 (iptables module(s) loaded, but no rules active)
Suggestions reviewed	15 (the first 15 listed in the report), including DEB-0880, AUTH-9286, AUTH-9262, AUTH-9282, AUTH-9230, BOOT-5122, DEB-0280 and a note that the Lynis release is over 4 months old
Report and log locations	/var/log/lynis-report.dat and /var/log/lynis.log

Selected findings

ID	Risk / Control Issue And Evidence	Likely Owner	Priority	Recommended Remediation	Status
W4-F02	No active host firewall rules (FIRE-4512). The host does not filter inbound or outbound traffic.	System Administrator / Network Security	High	Enable ufw or nftables with default-deny inbound; allow only required services	Open
W4-F03	Password policy not configured: no min/max age (AUTH-9286), no strength module (AUTH-9262), no expiry dates (AUTH-9282), no hashing-round setting (AUTH-9230).	IAM / Security Administrator	Moderate	Set aging in /etc/login.defs; install libpam-passwdqc	Open
W4-F04	No automated response to repeated authentication failures (DEB-0880: fail2ban not installed). Linked to the three logged failures in 4.2.	Security Operations / SysAdmin	Moderate	Install and configure fail2ban or a SIEM threshold alert	Open

Lower-priority observations: no GRUB boot password (BOOT-5122, low risk on a lab VM) and an outdated Lynis release (assessment-tool currency).

Additional findings from Bundles 1 and 2

ID	Condition and evidence	Likely owner	Priority	Status
W4-F01	/var/log/auth.log and /var/log/syslog do not exist. The auth_failures audit rule watches a non-existent file, so an empty ausearch result would appear to mean "no failures" when it means "no coverage".	System Administrator / Security Operations	Moderate	Open
W4-F05	auditd reports "No plugins found, not dispatching events": audit events are not forwarded to any central platform.	Security Operations	Moderate	Open

5. Control-Monitoring Table

Rows 1 to 6 are all based directly on evidence collected in this laboratory.

Control / Objective	Evidence	Owner	Observed Status	KPI/KRI or Threshold	Risk / Significance	Remediation	Retest / Follow-up
1. Audit logging of sensitive files and privileged use (accountability)	auditctl -l; ausearch/aureport: passwd_changes, shadow_changes, program_execution firing; auid=icdfa preserved under sudo	Security Operations / SysAdmin	Effective	100% of required rules loaded; any write to passwd/shadow outside a change window triggers an alert	Actions attributable to a named user; read-level watches are noisy (617 passwd events)	Keep rules; tune read events before SIEM forwarding	Monthly auditctl -l check and after every reboot
2. Authentication log coverage	/var/log/auth.log missing; auth_failures rule watches a non-existent file	SysAdmin / Security Operations	Gap: false assurance	100% of authentication sources monitored	Empty results would look like "no failures" when there is no coverage	Move to journal-based monitoring or enable a file-based auth log; update audit rule	Generate a benign failed login and confirm it is captured
3. Failed-authentication monitoring and response	Journal: 3 failures for icdfa (10 May, 20 Sep, 24 Sep); Lynis DEB-0880	Security Operations	Partially effective: logged, not acted on	Alert at >5 failures per account in 10 min; escalate if privileged/root account	Isolated, likely mistyped, but no automated response exists	Install and configure fail2ban or a SIEM alert rule	Simulated failures trigger an alert or ban
4. Host firewall (network protection)	Lynis warning FIRE-4512: iptables loaded, no rules active	SysAdmin / Network Security	Not effective	0 hosts without an active default-deny policy	Unfiltered inbound and outbound traffic	Enable ufw or nftables, default-deny, allow required services only	Re-run Lynis; FIRE-4512 should clear; check ufw status
5. Password policy (credential strength)	Lynis AUTH-9286, AUTH-9262, AUTH-9282, AUTH-9230	IAM / Security Administrator	Partially implemented	100% of accounts with max age per policy (e.g. 90 days) and strength enforcement	Weak or non-expiring passwords raise compromise risk	Set aging in /etc/login.defs; install libpam-passwdqc	Re-run Lynis; verify with chage -l
6. Centralised log forwarding	auditd startup: "No plugins found, not dispatching events"	Security Operations	Not implemented	100% of in-scope hosts forwarding audit and auth logs	Local-only logs can be lost or altered and are not correlated across hosts	Configure audit dispatcher or agent to forward to SIEM	Confirm test events appear in the SIEM within 5 min

6. SIEM / Continuous Monitoring Mapping

Linux Evidence	How it reaches the SIEM	SIEM use	Technical alert (SOC)	Governance / control issue
auditd events	Audit dispatcher plugin or agent (e.g. Wazuh or Splunk forwarder)	Correlate user, command and file activity	Write to shadow/passwd; auditd stopped	Audit coverage gaps; rules not loaded
journald (auth, sudo, errors)	Journal export or rsyslog / agent	Failed-login trends; privileged-use reports	>5 failures in 10 min; unusual sudo user	Authentication source missing from monitoring
Lynis report	Scheduled parse of /var/log/lynis-report.dat	Dashboard of hardening index, warnings over time	n/a (periodic, not real-time)	Index below target; overdue findings
Firewall / service state	Scheduled script or agent check	Continuous control testing	Firewall disabled unexpectedly	Control absent on multiple hosts

Governance discussion

Host-level evidence can be forwarded to a SIEM through agents, the audit dispatcher or journal export, then normalised and correlated across systems. In this lab, auditd reported that no plugins were configured, so forwarding would need to be set up first. Time-sensitive events (repeated authentication failures, writes to account files, auditd stopping) should remain technical operational alerts handled by the SOC. Persistent conditions that show a control is missing or ineffective (no active firewall, missing authentication log coverage, a low or declining hardening index, overdue remediation) should become governance issues, tracked with an owner, threshold and retest date, and reported to management through dashboards and recurring control-assurance reports.

Continuous control monitoring could automate the checks used here: scheduled verification that auditd is active and rules are loaded, automated Lynis scans with trend reporting, alerting on authentication thresholds, and workflow tickets that assign and track remediation to closure.

7. Governance Escalation Questions

Highest priority finding: W4-F02, the missing host firewall. It is the only Lynis warning and a preventive control is entirely absent. W4-F01 follows because it undermines detection.

Remediation owners: firewall: System Administrator / Network Security; password policy: IAM; authentication monitoring and fail2ban: Security Operations; audit coverage and log forwarding: Security Operations with the System Administrator.

Escalation triggers: escalate to the CISO if a High finding stays open beyond 30 days, any write to /etc/shadow or /etc/passwd occurs outside a change window, auditd stops, or a privileged account has more than 5 failures in 10 minutes. Escalate to the Risk function or management committee if a critical control gap affects several systems, a remediation deadline is missed twice, or the hardening index falls below the target (e.g. 70).

Evidence of successful remediation: a Lynis re-scan with FIRE-4512 cleared and a higher index; ufw status showing active rules; chage -l showing the new password aging; a simulated failed login that triggers an alert or ban; a journal-based authentication source confirmed in monitoring; test events visible in the SIEM.

Retest: after each fix, re-run the specific check or Lynis; then monthly for auditd rules and firewall status, and quarterly for a full Lynis scan, with results logged in the control register.

8. Remediation and Retest Plan

ID	Action	Owner	Priority	Retest evidence
W4-F02	Enable host firewall with default-deny inbound	SysAdmin / Network Security	High (30 days)	Lynis re-scan clears FIRE-4512; ufw status output
W4-F01	Enable journal-based auth monitoring or a file-based auth log; update audit rule	SysAdmin / SecOps	Moderate (60 days)	Benign failed login captured in the chosen source
W4-F03	Configure password aging and strength enforcement	IAM	Moderate (60 days)	chage -l output; Lynis AUTH findings cleared
W4-F04	Deploy fail2ban or SIEM threshold alert	Security Operations	Moderate (60 days)	Simulated failures trigger an alert or ban
W4-F05	Configure audit event forwarding to SIEM	Security Operations	Moderate (90 days)	Test events appear in the SIEM within 5 min

Timelines are recommended targets for management approval, not requirements set by the course.

9. Conclusion

The assigned Linux system has working detective controls: auditd is running, custom rules are loaded, and privileged and credential-file activity is attributable to a named user. No evidence of compromise was found. However, preventive and response controls are incomplete. There is no active firewall, password policy is not configured, authentication logging depends on the journal while an audit rule targets a missing file, and audit events are not forwarded to a central platform. Addressing the findings in the remediation plan, and retesting on a regular schedule, would move the control posture from partially effective to effective and give management dependable, consistent assurance evidence.

Appendix A: Supporting Evidence

Insert screenshots and command output under the headings below, in this order.

A1. auditd service status (systemctl status auditd) and custom.rules file in nano

```

(icdfa@icdfa)-[~]
$ sudo systemctl start auditd
sudo systemctl enable auditd

Synchronizing state of auditd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable auditd
Created symlink '/etc/systemd/system/multi-user.target.wants/auditd.service' -> '/usr/lib/systemd/system/auditd.service'.

(icdfa@icdfa)-[~]
$ sudo systemctl status auditd
● auditd.service - Security Audit Logging Service
   Loaded: loaded (/usr/lib/systemd/system/auditd.service; enabled; prese>
   Active: active (running) since Thu 2026-10-01 23:37:54 WAT; 15s ago
  Invocation: 99963610cc5e4a3382c8ba8111efc68a
     Docs: man:auditd(8)
           https://github.com/linux-audit/audit-documentation
    Main PID: 14369 (auditd)
      Tasks: 2 (limit: 2207)
     Memory: 668K (peak: 1.8M)
  
```

```

GNU nano 8.3 /etc/audit/rules.d/custom.rules
#Add rules that monitor sensitive account files
-w /etc/passwd -p rwx -k passwd_changes
-w /etc/shadow -p rwx -k shadow_changes

#Add program-execution monitoring rules
-a always,exit -F arch=b64 -S execve -k program_execution
-a always,exit -F arch=b32 -S execve -k program_execution

#Where /var/log/auth.log exists, add a watch for authentication-log changes
-w /var/log/auth.log -p wa -k auth_failures

Oct 01 23:37:54 icdfa systemd[1]: Starting auditd.service - Security Audit
Oct 01 23:37:54 icdfa auditd[14369]: No plugins found, not dispatching even>
Oct 01 23:37:54 icdfa auditd[14369]: Init complete, auditd 4.1.2 listening
Oct 01 23:37:54 icdfa systemd[1]: Started auditd.service - Security Audit
  
```

[Read 10 lines]

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify	^_ Go To Line

A2. auditctl -l output showing the five loaded rules

```

(icdfa@icdfa)-[~]
$ sudo nano /etc/audit/rules.d/custom.rules

(icdfa@icdfa)-[~]
$ sudo systemctl restart auditd
sudo auditctl -l

-w /etc/passwd -p rwx -k passwd_changes
-w /etc/shadow -p rwx -k shadow_changes
-a always,exit -F arch=b64 -S execve -F key=program_execution
-a always,exit -F arch=b32 -S execve -F key=program_execution
-w /var/log/auth.log -p wa -k auth_failures

(icdfa@icdfa)-[~]
$

```

A3. ausearch results (passwd_changes, shadow_changes, program_execution) with date

```

systemd-private-17bb5a84f11a4903976728af398ff6ea-ModemManager.service-81fYDz
systemd-private-17bb5a84f11a4903976728af398ff6ea-polkit.service-NnHquh
systemd-private-17bb5a84f11a4903976728af398ff6ea-systemd-logind.service-cynkog
systemd-private-17bb5a84f11a4903976728af398ff6ea-upower.service-9ERZ84
VMwareDnD
vmware-root_674-2731152261
(icdfa@icdfa)-[~]
$ sudo ausearch -k passwd_changes
sudo ausearch -k program_execution | tail -30
sudo ausearch -k auth_failures

time→Thu Oct  1 23:48:23 2026
type=PROCTITLE msg=audit(1790894903.859:164): proctitle=2F7362696E2F61756469746374
6C002D52002F6574632F61756469742F61756469742E72756C6573
type=SYSCALL msg=audit(1790894903.859:164): arch=c000003e syscall=44 success=yes e
xit=1084 a0=3 a1=7ffe28574c10 a2=43c a3=0 items=0 ppid=19959 pid=19984 auid=429496
7295 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 fsgid=0 tty=(none) ses=429496
7295 comm="auditctl" exe="/usr/sbin/auditctl" subj=unconfined key=(null)
type=CONFIG_CHANGE msg=audit(1790894903.859:164): auid=4294967295 ses=4294967295 s
ubj=unconfined op=add_rule key="passwd_changes" list=4 res=1

```

```
icdfa@icdfa: ~  
File Actions Edit View Help  
icdfa@icdfa: ~  
$ sudo aureport  
sudo aureport --failed  
sudo aureport --login  
Summary Report  
Range of time in logs: 01/10/2026 23:37:54.947 - 02/10/2026 00:18:16.199  
Selected time for report: 01/10/2026 23:37:54 - 02/10/2026 00:18:16.199  
Number of changes in configuration: 24  
Number of changes to accounts, groups, or roles: 0  
Number of logins: 0  
Number of failed logins: 0  
Number of authentications: 1  
Number of failed authentications: 0  
Number of users: 3  
Number of terminals: 9  
Number of host names: 2  
Number of executables: 73  
Number of commands: 102  
Number of files: 140
```

```

Number of events: 14167
Security Audit Logging Service
Loaded: loaded (/usr/lib/systemd/system/auditd.service; enabled; prese
Active: active (running) since Thu 2026-10-01 23:37:54 WAT; 15s ago
Failed Summary Report 0ccc5e4a3382c8ba8111efc68a
=====
Range of time in logs: 01/10/2026 23:37:54.947 - 02/10/2026 00:18:16.479
Selected time for report: 01/10/2026 23:37:54 - 02/10/2026 00:18:16.479
Number of changes in configuration: 0
Number of changes to accounts, groups, or roles: 0
Number of logins: 0
Number of failed logins: 0
Number of authentications: 0
Number of failed authentications: 0
Number of users: 3
Number of terminals: 2
Number of host names: 0
Number of executables: 3
Number of commands: 3
Number of files: 6
Number of AVC's: 0
Number of MAC events: 0
Number of failed syscalls: 27

```

```

Number of AVC's: 0
Number of MAC events: 0
Number of failed syscalls: 27
Number of anomaly events: 0
Number of responses to anomaly events: 0
Number of crypto events: 0
Number of integrity events: 0
Number of virt events: 0
Number of keys: 2
Number of process IDs: 13
Number of events: 27

Login Report
=====
# date time audit host term exe success event
=====
<no events of interest were found>

(icdfa@icdfa)-[~]
$

```

A5. journalctl --since "today", -p err and _COMM=sudo output with date

```
File Actions Edit View Help
(icdfa@icdfa)-[~] Security Audit Logging Service
$ sudo journalctl
sudo journalctl -u ssh
sudo journalctl --since "today"
sudo journalctl -p err

Evidence of a recent system log review
shows the errors

May 10 20:50:54 icdfa kernel: Linux version 6.12.13-amd64 (devel@kal>
May 10 20:50:54 icdfa kernel: Command line: BOOT_IMAGE=/boot/vmlinuz>
May 10 20:50:54 icdfa kernel: BIOS-provided physical RAM map:
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x0000000000000000-0x0>
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x0000000000009fc00-0x0>
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000000f0000-0x0>
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x0000000000100000-0x0>
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x0000000007fff0000-0x0>
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fec00000-0x0>
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fee00000-0x0>
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fffc0000-0x0>
May 10 20:50:54 icdfa kernel: NX (Execute Disable) protection: active
May 10 20:50:54 icdfa kernel: APIC: Static calls initialized
May 10 20:50:54 icdfa kernel: SMBIOS 2.5 present.
May 10 20:50:54 icdfa kernel: DMI: innotek GmbH VirtualBox/VirtualBo>
May 10 20:50:54 icdfa kernel: DMI: Memory slots populated: 0/0
```

```
File Actions Edit View Help
auditd.service - Security Audit Logging Service
(icdfa@icdfa)-[~] (/usr/lib/systemd/system/auditd.service; enabled
$ date
sudo journalctl --no-pager --since "today" | tail -30
sudo journalctl --no-pager -p err | tail -20
sudo journalctl --no-pager _COMM=sudo | tail -15
Fri 02 Oct 2026 00:56:02 WAT
Oct 02 00:39:43 icdfa systemd[1]: Starting phpsessionclean.service - Clean php ses
sion files ...
Oct 02 00:39:44 icdfa systemd[1]: phpsessionclean.service: Deactivated successfull
y.
Oct 02 00:39:44 icdfa systemd[1]: Finished phpsessionclean.service - Clean php ses
sion files.
Oct 02 00:41:29 icdfa auditd[30139]: Audit daemon rotating log files
Oct 02 00:45:01 icdfa CRON[116049]: pam_unix(cron:session): session opened for use
r root(uid=0) by root(uid=0)
Oct 02 00:45:01 icdfa CRON[116051]: (root) CMD (command -v debian-sa1 > /dev/null
&& debian-sa1 1 1)
Oct 02 00:45:01 icdfa CRON[116049]: pam_unix(cron:session): session closed for use
r root
Oct 02 00:46:26 icdfa sudo[116738]: icdfa : TTY=pts/2 ; PWD=/home/icdfa ; USER=
root ; COMMAND=/usr/bin/journalctl
```

File Actions Edit View Help

```

auditd.service - Security Audit Logging Service
└─(icdfa@icdfa)-[~] /usr/lib/systemd/system/auditd.service; enabled
└─$ sudo journalctl -f
Oct 02 01:05:44 icdfa sudo[291877]: pam_unix(sudo:session): session opened for use
r root(uid=0) by icdfa(uid=1000)
Oct 02 01:05:44 icdfa sudo[291877]: pam_unix(sudo:session): session closed for use
r root
Oct 02 01:05:44 icdfa sudo[291909]:      icdfa : TTY=pts/2 ; PWD=/home/icdfa ; USER=
root ; COMMAND=/usr/bin/journalctl --since today
Oct 02 01:05:44 icdfa sudo[291909]: pam_unix(sudo:session): session opened for use
r root(uid=0) by icdfa(uid=1000)
Oct 02 01:05:46 icdfa sudo[291909]: pam_unix(sudo:session): session closed for use
r root
Oct 02 01:05:46 icdfa sudo[293258]:      icdfa : TTY=pts/2 ; PWD=/home/icdfa ; USER=
root ; COMMAND=/usr/bin/journalctl -p err
Oct 02 01:05:46 icdfa sudo[293258]: pam_unix(sudo:session): session opened for use
r root(uid=0) by icdfa(uid=1000)
Oct 02 01:05:48 icdfa sudo[293258]: pam_unix(sudo:session): session closed for use
r root
Oct 02 01:05:54 icdfa sudo[297242]:      icdfa : TTY=pts/2 ; PWD=/home/icdfa ; USER=
root ; COMMAND=/usr/bin/journalctl -f
Oct 02 01:05:54 icdfa sudo[297242]: pam_unix(sudo:session): session opened for use

```

File Actions Edit View Help

```

Oct 02 01:05:54 icdfa sudo[297242]: pam_unix(sudo:session): session opened for use
r root(uid=0) by icdfa(uid=1000)
Oct 02 01:06:02 icdfa auditd[30139]: Audit daemon rotating log files
Oct 02 01:06:21 icdfa NetworkManager[786]: <info> [1790899581.9316] dhcp4 (eth0):
state changed new lease, address=10.10.1.137
Oct 02 01:06:21 icdfa dbus-daemon[670]: [system] Activating via systemd: service n
ame='org.freedesktop.nm_dispatcher' unit='dbus-org.freedesktop.nm-dispatcher.servi
ce' requested by ':1.5' (uid=0 pid=786 comm="/usr/sbin/NetworkManager --no-daemon"
)
Oct 02 01:06:21 icdfa systemd[1]: Starting NetworkManager-dispatcher.service - Net
work Manager Script Dispatcher Service ...
Oct 02 01:06:22 icdfa dbus-daemon[670]: [system] Successfully activated service 'o
rg.freedesktop.nm_dispatcher'
Oct 02 01:06:22 icdfa systemd[1]: Started NetworkManager-dispatcher.service - Netw
ork Manager Script Dispatcher Service.
Oct 02 01:06:32 icdfa systemd[1]: NetworkManager-dispatcher.service: Deactivated s
uccessfully.
Oct 02 01:06:34 icdfa auditd[30139]: Audit daemon rotating log files
Oct 02 01:06:57 icdfa auditd[30139]: Audit daemon rotating log files
Oct 02 01:07:20 icdfa auditd[30139]: Audit daemon rotating log files
Oct 02 01:07:40 icdfa auditd[30139]: Audit daemon rotating log files

```

A6. Authentication checks: missing /var/log/auth.log, journal search for failed password and authentication failure

```
(icdfa@icdfa)-[~]
$ sudo less /var/log/auth.log
sudo grep -i "failed password" /var/log/auth.log
sudo grep -i "sudo" /var/log/auth.log

/var/log/auth.log: No such file or directory
grep: /var/log/auth.log: No such file or directory
grep: /var/log/auth.log: No such file or directory

(icdfa@icdfa)-[~]
$
```

A7. Warning review: missing /var/log/syslog and journalctl -p warning output

```
File Actions Edit View Help
/var/log/auth.log: No such file or directory
grep: /var/log/auth.log: No such file or directory
grep: /var/log/auth.log: No such file or directory

(icdfa@icdfa)-[~]
$ date
sudo journalctl --no-pager | grep -i "failed password" | tail -5
sudo journalctl --no-pager | grep -i "authentication failure" | tail -5
sudo journalctl --no-pager | grep -i "sudo" | tail -15
Fri 02 Oct 2026 01:15:20 WAT
Oct 02 01:10:10 icdfa sudo[422706]: icdfa : TTY=pts/2 ; PWD=/home/icdfa ; USER=
root ; COMMAND=/usr/bin/grep -i 'failed password' /var/log/auth.log
May 10 21:20:11 icdfa sudo[13196]: pam_unix(sudo:auth): authentication failure; lo
gname=icdfa uid=1000 euid=0 tty=/dev/pts/0 ruser=icdfa rhost= user=icdfa
Sep 20 02:47:01 icdfa polkit-agent-helper-1[43340]: pam_unix(polkit-1:auth): authe
ntication failure; logname=icdfa uid=1000 euid=0 tty= ruser=icdfa rhost= user=icd
fa
Sep 24 20:09:02 icdfa lightdm[1837]: pam_unix(lightdm:auth): authentication failur
e; logname= uid=0 euid=0 tty=:0 ruser= rhost= user=icdfa
Oct 02 01:10:10 icdfa sudo[422693]: pam_unix(sudo:session): session closed for use
r root
```

File Actions Edit View Help

```

r root(uid=0) by icdfa(uid=1000)
Oct 02 01:10:10 icdfa sudo[422722]: pam_unix(sudo:session): session closed for use
r root
Oct 02 01:15:20 icdfa sudo[568459]:    icdfa : TTY=pts/2 ; PWD=/home/icdfa ; USER=
root ; COMMAND=/usr/bin/journalctl --no-pager
Oct 02 01:15:20 icdfa sudo[568459]: pam_unix(sudo:session): session opened for use
r root(uid=0) by icdfa(uid=1000)
Oct 02 01:15:22 icdfa sudo[568459]: pam_unix(sudo:session): session closed for use
r root
Oct 02 01:15:22 icdfa sudo[569103]:    icdfa : TTY=pts/2 ; PWD=/home/icdfa ; USER=
root ; COMMAND=/usr/bin/journalctl --no-pager
Oct 02 01:15:22 icdfa sudo[569103]: pam_unix(sudo:session): session opened for use
r root(uid=0) by icdfa(uid=1000)
Oct 02 01:15:24 icdfa sudo[569103]: pam_unix(sudo:session): session closed for use
r root
Oct 02 01:15:24 icdfa sudo[569624]:    icdfa : TTY=pts/2 ; PWD=/home/icdfa ; USER=
root ; COMMAND=/usr/bin/journalctl --no-pager
Oct 02 01:15:24 icdfa sudo[569624]: pam_unix(sudo:session): session opened for use
r root(uid=0) by icdfa(uid=1000)

```

(icdfa@icdfa)-[~]

\$

```

(icdfa@icdfa)-[~] (running) since Thu 2026-10-01 23:37:54 WAT; 15s
$ date
ls -l /var/log/syslog
sudo journalctl --no-pager -p warning | tail -15
Fri 02 Oct 2026 01:22:23 WAT
ls: cannot access '/var/log/syslog': No such file or directory
Oct 01 23:22:44 icdfa systemd-sysv-generator[6912]: ! This compatibility logic is
deprecated, expect removal soon. !
Oct 01 23:22:45 icdfa systemd-sysv-generator[6995]: SysV service '/etc/init.d/inet
sim' lacks a native systemd unit file, automatically generating a unit file for co
mpatibility.
Oct 01 23:22:45 icdfa systemd-sysv-generator[6995]: Please update package to inclu
de a native systemd unit file.
Oct 01 23:22:45 icdfa systemd-sysv-generator[6995]: ! This compatibility logic is
deprecated, expect removal soon. !
Oct 01 23:36:09 icdfa rtkit-daemon[1537]: The canary thread is apparently starving
. Taking action.
Oct 01 23:37:55 icdfa systemd-sysv-generator[14469]: SysV service '/etc/init.d/ine
tsim' lacks a native systemd unit file, automatically generating a unit file for c
ompatibility.

```

A8. Lynis 3.1.4 completion screen, hardening index, warning, suggestions and report_datetime_end

```

(icdfa@icdfa)-[~]
$ date
lynis --version
Fri 02 Oct 2026 01:41:08 WAT
3.1.4

```

(icdfa@icdfa)-[~]

\$

```
File Actions Edit View Help
(icdfa@icdfa)-[~]
$ sudo lynis audit system

[ Lynis 3.1.4 ]

#####
###
Lynis comes with ABSOLUTELY NO WARRANTY. This is free software, and you are
welcome to redistribute it under the terms of the GNU General Public License.
See the LICENSE file for details about using this software.

2007-2024, CISofy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)
#####
###

[+] Initializing program
-----
- Detecting OS ... [ DONE ]
- Checking profiles ... [ DONE ]
```

```
File Actions Edit View Help
- Report data : /var/log/lynis-report.dat

=====
Lynis 3.1.4

Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)

2007-2024, CISofy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)

=====

[TIP]: Enhance Lynis audits by adding your settings to custom.prf (see /etc
/lynis/default.prf for all settings)

(icdfa@icdfa)-[~]
$
```

```
icdfa@icdfa: ~  
File Actions Edit View Help  
Home DELAB  
2007-2024, CISOfy - https://cisofy.com/lynis/  
Enterprise support available (compliance, plugins, interface and tools)  
  
[TIP]: Enhance Lynis audits by adding your settings to custom.prf (see /etc  
/lynis/default.prf for all settings)  
  
(icdfa@icdfa)-[~]  
$ date  
sudo grep "hardening_index" /var/log/lynis-report.dat  
sudo grep "^warning\[\\]" /var/log/lynis-report.dat  
Fri 02 Oct 2026 03:37:09 WAT  
[sudo] password for icdfa:  
hardening_index=65  
warning[]=FIRE-4512|iptables module(s) loaded, but no rules active  
  
(icdfa@icdfa)-[~]  
$
```

```
(icdfa@icdfa)-[~]  
$ sudo grep "report_datetime_end" /var/log/lynis-report.dat  
report_datetime_end=2026-10-02 03:25:35  
  
(icdfa@icdfa)-[~]  
$
```

The screenshot shows a terminal window with the nano text editor open to the file `/etc/passwd`. The file contains the following user entries:

```

root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin

```

The nano editor interface shows the file name `/etc/passwd` in the title bar and the GNU nano 8.3 version. The bottom status bar displays various keyboard shortcuts for editing and navigation.

The screenshot shows a terminal window with the following commands and output:

```

(icdfa@icdfa)-[~] /usr/lib/systemd/system/auditd.service; enabled; prese
$ sudo nano /etc/audit/rules.d/custom.rules
(icdfa@icdfa)-[~]
$ sudo nano /etc/passwd
(icdfa@icdfa)-[~]
$ ls /tmp
config-err-dx0FPA
ssh-M1MOu8U2R5JA
systemd-private-17bb5a84f11a4903976728af398ff6ea-colord.service-woHr0x
systemd-private-17bb5a84f11a4903976728af398ff6ea-haveged.service-cFDXaC
systemd-private-17bb5a84f11a4903976728af398ff6ea-ModemManager.service-81fYDz
systemd-private-17bb5a84f11a4903976728af398ff6ea-polkit.service-NnHquh
systemd-private-17bb5a84f11a4903976728af398ff6ea-systemd-logind.service-cynkog
systemd-private-17bb5a84f11a4903976728af398ff6ea-upower.service-9ERZ84
VMwareDnD
vmware-root_674-2731152261

```

The terminal shows the user `icdfa` at the `icdfa` host. The commands executed are `sudo nano /etc/audit/rules.d/custom.rules`, `sudo nano /etc/passwd`, and `ls /tmp`. The output of `ls /tmp` lists various system service files and directories in the `/tmp` directory.

The use of AI Declaration

I used an AI assistant (Claude, developed by Anthropic) while completing the lab assignment and report. The assistance was limited to the following:

Explanation: clarifying what commands do and how to read auditd, journalctl and Lynis output.

Report structuring: suggesting the layout of the report, the control-monitoring table and the SIEM mapping.

Language improvement: drafting and refining the wording of the interpretations, findings and conclusion, based on the evidence I supplied.

I personally performed all required lab activities in my assigned ICDFA virtual machine, including configuring auditd rules, generating the audit events, running the log-analysis commands and running the Lynis audit.

All command outputs, screenshots and timestamps come from my own authorised ICDDA lab work, and nothing was fabricated. I reviewed the AI-assisted text against my evidence and corrected it where needed.

INTERNATIONAL CYBERSECURITY AND
DIGITAL FORENSICS ACADEMY

Week 4 Practical Laboratory

Linux Security Monitoring and Auditing

From Technical Evidence to Governance Assurance

Name	Adeshola Oluwole
Registration Number	C1126CGRCE17211
Course	GRC102 – Information Security Governance
Module	Module 4 – Monitoring and Auditing Security Controls
Week	Week 4: 26 September – 02 October 2026
Opens	26 September 2026, 8:00 AM WAT
Due	02 October 2026, 11:59 PM WAT
Maximum Score	100 marks
Gradebook Contribution	10% of final course grade (25% within Practical Laboratories)
Level / Mode	Intermediate • Self-Paced • Individual practical evidence

Submission mode: Written report + supporting evidence

Platform: ICDFA Ubuntu Practice Hub / authorised assigned Linux VM

1. Laboratory Overview

This laboratory provides hands-on experience in Linux security monitoring and auditing. You will configure and query the Linux auditing subsystem, review authentication and system logs, perform a system security assessment, and translate technical findings into a governance-focused control-monitoring and audit report.

The lab uses Linux command-line tools because governance and assurance decisions depend on reliable evidence. Your objective is not merely to run commands: you must explain what the evidence means, identify conditions requiring governance attention, assign control ownership, and recommend remediation and follow-up.

2. Learning Objectives

- Understand the role of auditd and configure audit rules for selected security-relevant activities.
- Use ausearch and aureport to retrieve and summarise Linux audit evidence.
- Use journalctl, grep and traditional Linux log files to investigate authentication, privilege-use and system events.
- Perform a Linux security assessment using Lynis and interpret warnings, suggestions and hardening indicators.
- Identify technical conditions that should be escalated through security governance or control-assurance processes.
- Connect Linux evidence to control objectives, control owners, status, remediation actions and retesting requirements.
- Explain conceptually how SIEM and automation extend host-level monitoring into continuous control monitoring.

3. Authorised Lab Rules and Prerequisites

Complete all hands-on activities only inside the Linux system assigned to you for training. Do not target external systems, production infrastructure or any device for which you do not have explicit authorisation.

- Use the ICDFA Ubuntu Practice Hub, and Metasploitable VM, or another instructor-authorised isolated Linux environment.
- Basic Linux CLI familiarity is required, including cd, ls, sudo and a text editor such as nano or vim.
- Internet access may be required to install auditd or Lynis if they are not already available.
- Preserve command output, screenshots or exported text that demonstrates completion of each required activity.
- Do not intentionally damage the operating system, delete evidence, disable logging, or make unrelated configuration changes.
- Where a distribution stores logs differently, document the equivalent log source used.

4. Scenario Governance Assurance for a Linux Workload

Your organisation relies on Linux systems for business-critical services. Management has approved security controls for authentication monitoring, privileged access, system auditing, vulnerability and configuration assessment, but internal assurance reviews show that evidence is collected inconsistently.

You are acting as a Security Control Assurance Analyst. Your task is to inspect an assigned Linux system, generate and analyse security evidence, identify issues or events requiring governance attention, and produce a control-monitoring record and concise audit report for management.

5. Module 1 System Auditing with auditd

auditd is the userspace component of the Linux Auditing System. It records security-relevant events such as file access, system calls and user actions, providing evidence useful for forensic analysis, compliance and continuous monitoring.

Activity 1.1 Verify or Install auditd

Check whether the audit service is present and active:

```
sudo systemctl status auditd
```

If auditd is not installed, install and enable it in your assigned lab VM:

```
sudo apt update
sudo apt install auditd audispd-plugins
sudo systemctl start auditd
sudo systemctl enable auditd
```

Re-check the service status and record evidence showing whether it is active.

The screenshot shows a terminal window with the following commands and output:

```
(icdfa@icdfa)-[~]
$ sudo systemctl start auditd
$ sudo systemctl enable auditd
```

Synchronizing state of auditd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
 Executing: /usr/lib/systemd/systemd-sysv-install enable auditd
 Created symlink '/etc/systemd/system/multi-user.target.wants/auditd.service' → '/usr/lib/systemd/system/auditd.service'.

```
(icdfa@icdfa)-[~]
$ sudo systemctl status auditd
```

```
● auditd.service - Security Audit Logging Service
   Loaded: loaded (/usr/lib/systemd/system/auditd.service; enabled; prese>
   Active: active (running) since Thu 2026-10-01 23:37:54 WAT; 15s ago
   Invocation: 99963610cc5e4a3382c8ba8111efc68a
   Docs: man:auditd(8)
        https://github.com/linux-audit/audit-documentation
   Main PID: 14369 (auditd)
   Tasks: 2 (limit: 2207)
   Memory: 668K (peak: 1.8M)
```

Activity 1.2 Configure Audit Rules

Create a dedicated rules file rather than editing the primary rules file directly:

```
sudo nano /etc/audit/rules.d/custom.rules
```

Add rules that monitor sensitive account files:

```
-w /etc/passwd -p rwx -k passwd_changes
-w /etc/shadow -p rwx -k shadow_changes
```

Add program-execution monitoring rules:

```
-a always,exit -F arch=b64 -S execve -k program_execution
-a always,exit -F arch=b32 -S execve -k program_execution
```

Where /var/log/auth.log exists, add a watch for authentication-log changes:

```
-w /var/log/auth.log -p wa -k auth_failures
```

Save the file, then load or restart the audit service as supported by your VM:

```
sudo systemctl restart auditd
sudo auditctl -l
```

Record the loaded rules. If your distribution restricts manual auditd restarts, document the approved equivalent method used in your assigned environment.

```
GNU nano 8.3 ce - Security/etc/audit/rules.d/custom.rules
#Add rules that monitor sensitive account files
-w /etc/passwd -p rwx -k passwd_changes
-w /etc/shadow -p rwx -k shadow_changes

#Add program-execution monitoring rules
-a always,exit -F arch=b64 -S execve -k program_execution
-a always,exit -F arch=b32 -S execve -k program_execution

#Where /var/log/auth.log exists, add a watch for authentication-log changes
-w /var/log/auth.log -p wa -k auth_failures

Oct 01 23:37:54 icdfa systemd[1]: Starting auditd.service - Security Audit
Oct 01 23:37:54 icdfa auditd[14369]: No plugins found, not dispatching even
Oct 01 23:37:54 icdfa auditd[14369]: Init complete, auditd 4.1.2 listening
Oct 01 23:37:54 icdfa systemd[1]: Started auditd.service - Security Audit
```

[Read 10 lines]

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify	^_ Go To Line

```

(icdfa@icdfa)-[~]
$ sudo nano /etc/audit/rules.d/custom.rules

(icdfa@icdfa)-[~]
$ sudo systemctl restart auditd
sudo auditctl -l

-w /etc/passwd -p rwx -k passwd_changes
-w /etc/shadow -p rwx -k shadow_changes
-a always,exit -F arch=b64 -S execve -F key=program_execution
-a always,exit -F arch=b32 -S execve -F key=program_execution
-w /var/log/auth.log -p wa -k auth_failures

(icdfa@icdfa)-[~]
$

```

Activity 1.3 Generate and Query Audit Events

Generate only benign events inside your assigned VM. Do not save changes to sensitive account files.

```

sudo nano /etc/passwd
# Exit without saving.
ls /tmp

```

Optionally generate a failed local authentication attempt in the assigned VM using an instructor-approved method. Do not attempt authentication against external hosts or accounts.

The screenshot shows a terminal window with the nano text editor open, displaying the contents of the `/etc/passwd` file. The file lists system and regular user accounts with their respective UID, GID, home directory, and shell. The `/etc/passwd` path is highlighted with a red box. The terminal window title is `icdfa@icdfa: ~`. The nano editor status bar at the bottom shows `[ Read 46 lines ]` and various keyboard shortcuts.

```

GNU nano 8.3 /etc/passwd
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
[ Read 46 lines ]
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line

```

```

icdfa@icdfa: ~
File Actions Edit View Help

(icdfa@icdfa)-[~] (/usr/lib/systemd/system/auditd.service; enabled; prese
$ sudo nano /etc/audit/rules.d/custom.rules
Invocation: 99963610cc5e4a3382c8ba8111efc68a
(icdfa@icdfa)-[~]
$ sudo nano /etc/passwd
Main PID: 14369 (auditd)
220K (peak: 1.8M)
(icdfa@icdfa)-[~]
$ ls /tmp
config-err-dx0FPA
ssh-M1MOu8U2R5JA
systemd-private-17bb5a84f11a4903976728af398ff6ea-color.service-woHr0x
systemd-private-17bb5a84f11a4903976728af398ff6ea-haveged.service-cFDXaC
systemd-private-17bb5a84f11a4903976728af398ff6ea-ModemManager.service-81fYDz
systemd-private-17bb5a84f11a4903976728af398ff6ea-polkit.service-NnHquh
systemd-private-17bb5a84f11a4903976728af398ff6ea-systemd-logind.service-cynkog
systemd-private-17bb5a84f11a4903976728af398ff6ea-upower.service-9ERZ84
VMwareDnD
vmware-root_674-2731152261
(icdfa@icdfa)-[~]
$

```

Query audit events by key:

```

sudo ausearch -k passwd_changes
sudo ausearch -k program_execution
sudo ausearch -k auth_failures

```

```

icdfa@icdfa: ~
File Actions Edit View Help

systemd-private-17bb5a84f11a4903976728af398ff6ea-ModemManager.service-81fYDz
systemd-private-17bb5a84f11a4903976728af398ff6ea-polkit.service-NnHquh
systemd-private-17bb5a84f11a4903976728af398ff6ea-systemd-logind.service-cynkog
systemd-private-17bb5a84f11a4903976728af398ff6ea-upower.service-9ERZ84
VMwareDnD
vmware-root_674-2731152261
(icdfa@icdfa)-[~]
$ sudo ausearch -k passwd_changes
sudo ausearch -k program_execution | tail -30
sudo ausearch -k auth_failures

time→Thu Oct 1 23:48:23 2026
type=PROCTITLE msg=audit(1790894903.859:164): proctitle=2F7362696E2F61756469746374
6C002D52002F6574632F61756469742F61756469742E72756C6573
type=SYSCALL msg=audit(1790894903.859:164): arch=c000003e syscall=44 success=yes e
xit=1084 a0=3 a1=7ffe28574c10 a2=43c a3=0 items=0 ppid=19959 pid=19984 auid=429496
7295 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=(none) ses=429496
7295 comm="auditctl" exe="/usr/sbin/auditctl" subj=unconfined key=(null)
type=CONFIG_CHANGE msg=audit(1790894903.859:164): auid=4294967295 ses=4294967295 s
ubj=unconfined op=add_rule key="passwd_changes" list=4 res=1

```

```
sudo aureport
sudo aureport --failed
sudo aureport --login
```

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```

Number of events: 14167
Security Audit Logging Service
Loaded: loaded (/usr/lib/systemd/system/auditd.service; enabled; prese
Active: active (running) since Thu 2026-10-01 23:37:54 WAT; 15s ago
Failed Summary Report 0ccc5e4a3382c8ba8111efc68a
=====
Range of time in logs: 01/10/2026 23:37:54.947 - 02/10/2026 00:18:16.479
Selected time for report: 01/10/2026 23:37:54 - 02/10/2026 00:18:16.479
Number of changes in configuration: 0
Number of changes to accounts, groups, or roles: 0
Number of logins: 0
Number of failed logins: 0
Number of authentications: 0
Number of failed authentications: 0
Number of users: 3
Number of terminals: 2
Number of host names: 0
Number of executables: 3
Number of commands: 3
Number of files: 6
Number of AVC's: 0
Number of MAC events: 0
Number of failed syscalls: 27

```

```

Number of AVC's: 0
Number of MAC events: 0
Number of failed syscalls: 27
Number of anomaly events: 0
Number of responses to anomaly events: 0
Number of crypto events: 0
Number of integrity events: 0
Number of virt events: 0
Number of keys: 2
Number of process IDs: 13
Number of events: 27

Login Report
=====
# date time audit host term exe success event
=====
<no events of interest were found>

(icdfa@icdfa)-[~]
$

```

Required Evidence Module 1

- Screenshot or text evidence that auditd is running.
- Evidence showing your loaded custom audit rules.
- At least one ausearch result for a configured audit key.
- At least one aureport summary.
- A short interpretation explaining what the observed audit record says about user/action/time and why the event could matter to security governance.

Short Interpretation of findings

The audit record shows that user **icdfa** (auid=icdfa, session 3, terminal pts2) used **sudo** to access sensitive account files. At **00:29:41 WAT on 02/10/2026**, sudo opened `/etc/passwd` read-only, and at **00:29:43 WAT** the password-check helper (`unix_chkpwd`) opened `/etc/shadow` read-only. Both actions succeeded, and no write or modification events were recorded, which is consistent with the test in which `/etc/passwd` was opened and closed without saving.

The record keeps the original user's identity (auid=icdfa) even though the process ran with root privileges (euid=root). This matters to security governance because it shows that access to credential files and privileged activity are logged and traceable to a named individual, which supports accountability, access-control and change-management assurance. It also lets management confirm that no unauthorised changes were made to user accounts.

6. Module 2 Log Management and Analysis

Linux systems generate authentication, service, kernel and application evidence through systemd journals and traditional log files. You will use `journalctl` and `grep` to identify security-relevant events and distinguish routine activity from conditions that warrant further investigation.

Activity 2.1 — Explore Logs with `journalctl`

```
sudo journalctl
sudo journalctl -u ssh
sudo journalctl --since "today"
sudo journalctl -p err
```

To observe live events:

```
sudo journalctl -f
```

While following the journal, perform only normal benign activity in your assigned VM, then stop the live view with `Ctrl+C`.

File Actions Edit View Help

```
(icdfa@icdfa)-[~] security Audit Logging Service
$ sudo journalctl (/usr/lib/systemd/system/auditd.service; enabled
sudo journalctl -u ssh
sudo journalctl --since "today"
sudo journalctl -p err
```

Evidence of a recent system log review

shows the errors

```
May 10 20:50:54 icdfa kernel: Linux version 6.12.13-amd64 (devel@kal>
May 10 20:50:54 icdfa kernel: Command line: BOOT_IMAGE=/boot/vmlinuz>
May 10 20:50:54 icdfa kernel: BIOS-provided physical RAM map:
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x0000000000000000-0x0>
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x0000000000009fc00-0x0>
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000000f0000-0x0>
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x0000000000100000-0x0>
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x0000000007fff0000-0x0>
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fec00000-0x0>
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fee00000-0x0>
May 10 20:50:54 icdfa kernel: BIOS-e820: [mem 0x00000000fffc0000-0x0>
May 10 20:50:54 icdfa kernel: NX (Execute Disable) protection: active
May 10 20:50:54 icdfa kernel: APIC: Static calls initialized
May 10 20:50:54 icdfa kernel: SMBIOS 2.5 present.
May 10 20:50:54 icdfa kernel: DMI: innotek GmbH VirtualBox/VirtualBo>
May 10 20:50:54 icdfa kernel: DMI: Memory slots populated: 0/0
```

File Actions Edit View Help

```
auditd.service - Security Audit Logging Service
(icdfa@icdfa)-[~] (/usr/lib/systemd/system/auditd.service; enabled
$ date
sudo journalctl --no-pager --since "today" | tail -30
sudo journalctl --no-pager -p err | tail -20
sudo journalctl --no-pager _COMM=sudo | tail -15
Fri 02 Oct 2026 00:56:02 WAT
Oct 02 00:39:43 icdfa systemd[1]: Starting phpsessionclean.service - Clean php ses
sion files ...
Oct 02 00:39:44 icdfa systemd[1]: phpsessionclean.service: Deactivated successfull
y.
Oct 02 00:39:44 icdfa systemd[1]: Finished phpsessionclean.service - Clean php ses
sion files.
Oct 02 00:41:29 icdfa auditd[30139]: Audit daemon rotating log files
Oct 02 00:45:01 icdfa CRON[116049]: pam_unix(cron:session): session opened for use
r root(uid=0) by root(uid=0)
Oct 02 00:45:01 icdfa CRON[116051]: (root) CMD (command -v debian-sa1 > /dev/null
&& debian-sa1 1 1)
Oct 02 00:45:01 icdfa CRON[116049]: pam_unix(cron:session): session closed for use
r root
Oct 02 00:46:26 icdfa sudo[116738]: icdfa : TTY=pts/2 ; PWD=/home/icdfa ; USER=
root ; COMMAND=/usr/bin/journalctl
```

File Actions Edit View Help

```

auditd.service - Security Audit Logging Service
└─(icdfa@icdfa)-[~] /usr/lib/systemd/system/auditd.service; enabled
└─$ sudo journalctl -f
Oct 02 01:05:44 icdfa sudo[291877]: pam_unix(sudo:session): session opened for use
r root(uid=0) by icdfa(uid=1000)
Oct 02 01:05:44 icdfa sudo[291877]: pam_unix(sudo:session): session closed for use
r root
Oct 02 01:05:44 icdfa sudo[291909]:      icdfa : TTY=pts/2 ; PWD=/home/icdfa ; USER=
root ; COMMAND=/usr/bin/journalctl --since today
Oct 02 01:05:44 icdfa sudo[291909]: pam_unix(sudo:session): session opened for use
r root(uid=0) by icdfa(uid=1000)
Oct 02 01:05:46 icdfa sudo[291909]: pam_unix(sudo:session): session closed for use
r root
Oct 02 01:05:46 icdfa sudo[293258]:      icdfa : TTY=pts/2 ; PWD=/home/icdfa ; USER=
root ; COMMAND=/usr/bin/journalctl -p err
Oct 02 01:05:46 icdfa sudo[293258]: pam_unix(sudo:session): session opened for use
r root(uid=0) by icdfa(uid=1000)
Oct 02 01:05:48 icdfa sudo[293258]: pam_unix(sudo:session): session closed for use
r root
Oct 02 01:05:54 icdfa sudo[297242]:      icdfa : TTY=pts/2 ; PWD=/home/icdfa ; USER=
root ; COMMAND=/usr/bin/journalctl -f
Oct 02 01:05:54 icdfa sudo[297242]: pam_unix(sudo:session): session opened for use

```

File Actions Edit View Help

```

Oct 02 01:05:54 icdfa sudo[297242]: pam_unix(sudo:session): session opened for use
r root(uid=0) by icdfa(uid=1000)
Oct 02 01:06:02 icdfa auditd[30139]: Audit daemon rotating log files
Oct 02 01:06:21 icdfa NetworkManager[786]: <info> [1790899581.9316] dhcp4 (eth0):
state changed new lease, address=10.10.1.137
Oct 02 01:06:21 icdfa dbus-daemon[670]: [system] Activating via systemd: service n
ame='org.freedesktop.nm_dispatcher' unit='dbus-org.freedesktop.nm-dispatcher.servi
ce' requested by ':1.5' (uid=0 pid=786 comm="/usr/sbin/NetworkManager --no-daemon"
)
Oct 02 01:06:21 icdfa systemd[1]: Starting NetworkManager-dispatcher.service - Net
work Manager Script Dispatcher Service ...
Oct 02 01:06:22 icdfa dbus-daemon[670]: [system] Successfully activated service 'o
rg.freedesktop.nm_dispatcher'
Oct 02 01:06:22 icdfa systemd[1]: Started NetworkManager-dispatcher.service - Netw
ork Manager Script Dispatcher Service.
Oct 02 01:06:32 icdfa systemd[1]: NetworkManager-dispatcher.service: Deactivated s
uccessfully.
Oct 02 01:06:34 icdfa auditd[30139]: Audit daemon rotating log files
Oct 02 01:06:57 icdfa auditd[30139]: Audit daemon rotating log files
Oct 02 01:07:20 icdfa auditd[30139]: Audit daemon rotating log files
Oct 02 01:07:40 icdfa auditd[30139]: Audit daemon rotating log files

```

Activity 2.2 Authentication and Privilege-Use Analysis

On Ubuntu/Debian systems that use /var/log/auth.log:

```
sudo less /var/log/auth.log
sudo grep -i "failed password" /var/log/auth.log
sudo grep -i "sudo" /var/log/auth.log
```

If your distribution does not use /var/log/auth.log, use the equivalent journal or authentication log source and document it.

```
(icdfa@icdfa)-[~]
$ sudo less /var/log/auth.log
sudo grep -i "failed password" /var/log/auth.log
sudo grep -i "sudo" /var/log/auth.log

/var/log/auth.log: No such file or directory
grep: /var/log/auth.log: No such file or directory
grep: /var/log/auth.log: No such file or directory

(icdfa@icdfa)-[~]
$
```

```
File Actions Edit View Help
# auditd.service - Security Audit Logging Service
/var/log/auth.log: No such file or directory
grep: /var/log/auth.log: No such file or directory
grep: /var/log/auth.log: No such file or directory

(icdfa@icdfa)-[~]
$ date
sudo journalctl --no-pager | grep -i "failed password" | tail -5
sudo journalctl --no-pager | grep -i "authentication failure" | tail -5
sudo journalctl --no-pager | grep -i "sudo" | tail -15
Fri 02 Oct 2026 01:15:20 WAT
Oct 02 01:10:10 icdfa sudo[422706]: icdfa : TTY=pts/2 ; PWD=/home/icdfa ; USER=
root ; COMMAND=/usr/bin/grep -i 'failed password' /var/log/auth.log
May 10 21:20:11 icdfa sudo[13196]: pam_unix(sudo:auth): authentication failure; lo
gname=icdfa uid=1000 euid=0 tty=/dev/pts/0 ruser=icdfa rhost= user=icdfa
Sep 20 02:47:01 icdfa polkit-agent-helper-1[43340]: pam_unix(polkit-1:auth): authe
ntication failure; logname=icdfa uid=1000 euid=0 tty= ruser=icdfa rhost= user=icd
fa
Sep 24 20:09:02 icdfa lightdm[1837]: pam_unix(lightdm:auth): authentication failur
e; logname= uid=0 euid=0 tty=:0 ruser= rhost= user=icdfa
Oct 02 01:10:10 icdfa sudo[422693]: pam_unix(sudo:session): session closed for use
r root
```

```
File Actions Edit View Help
r root(uid=0) by icdfa(uid=1000)
Oct 02 01:10:10 icdfa sudo[422722]: pam_unix(sudo:session): session closed for use
r root
Oct 02 01:15:20 icdfa sudo[568459]:    icdfa : TTY=pts/2 ; PWD=/home/icdfa ; USER=
root ; COMMAND=/usr/bin/journalctl --no-pager
Oct 02 01:15:20 icdfa sudo[568459]: pam_unix(sudo:session): session opened for use
r root(uid=0) by icdfa(uid=1000)
Oct 02 01:15:22 icdfa sudo[568459]: pam_unix(sudo:session): session closed for use
r root
Oct 02 01:15:22 icdfa sudo[569103]:    icdfa : TTY=pts/2 ; PWD=/home/icdfa ; USER=
root ; COMMAND=/usr/bin/journalctl --no-pager
Oct 02 01:15:22 icdfa sudo[569103]: pam_unix(sudo:session): session opened for use
r root(uid=0) by icdfa(uid=1000)
Oct 02 01:15:24 icdfa sudo[569103]: pam_unix(sudo:session): session closed for use
r root
Oct 02 01:15:24 icdfa sudo[569624]:    icdfa : TTY=pts/2 ; PWD=/home/icdfa ; USER=
root ; COMMAND=/usr/bin/journalctl --no-pager
Oct 02 01:15:24 icdfa sudo[569624]: pam_unix(sudo:session): session opened for use
r root(uid=0) by icdfa(uid=1000)

(icdfa@icdfa)-[~]
$
```

As /var/log/auth.log does not exist on this system, the systemd journal was used as the equivalent source. A search for "failed password" returned no genuine events (the only match was the analyst's own search command). A search for "authentication failure" returned three historical events for user icdfa: sudo on 10 May 2026, polkit on 20 Sep 2026 and lightdm on 24 Sep 2026. The isolated pattern suggests mistyped credentials and not a brute-force attempt, but it confirms that failed authentication is logged and attributable. Sudo activity on 02 Oct 2026 was recorded with user, terminal, command and time, supporting privileged-access accountability.

Activity 2.3 General System Log Review

Where /var/log/syslog exists:

```
sudo less /var/log/syslog
sudo grep -i "error" /var/log/syslog
sudo grep -i "warning" /var/log/syslog
```

If syslog is not present, use journalctl or the appropriate distribution-specific log source.

```
(icdfa@icdfa)-[~]
$ date
ls -l /var/log/syslog
sudo journalctl --no-pager -p warning | tail -15
Fri 02 Oct 2026 01:22:23 WAT
ls: cannot access '/var/log/syslog': No such file or directory
Oct 01 23:22:44 icdfa systemd-sysv-generator[6912]: ! This compatibility logic is
deprecated, expect removal soon. !
Oct 01 23:22:45 icdfa systemd-sysv-generator[6995]: SysV service '/etc/init.d/inet
sim' lacks a native systemd unit file, automatically generating a unit file for co
mpatibility.
Oct 01 23:22:45 icdfa systemd-sysv-generator[6995]: Please update package to inclu
de a native systemd unit file.
Oct 01 23:22:45 icdfa systemd-sysv-generator[6995]: ! This compatibility logic is
deprecated, expect removal soon. !
Oct 01 23:36:09 icdfa rtkit-daemon[1537]: The canary thread is apparently starving
. Taking action.
Oct 01 23:37:55 icdfa systemd-sysv-generator[14469]: SysV service '/etc/init.d/ine
tsim' lacks a native systemd unit file, automatically generating a unit file for c
ompatibility.
```

Required Evidence Module 2

- Evidence of recent system-log review using journalctl.
- Authentication or privilege-use evidence showing at least one relevant event or confirming that no significant event was observed.
- Error/warning evidence from syslog or an equivalent source.
- A short table containing Event/Condition, Evidence Source, Time, User/Service, Security Significance and Recommended Action.

Short Explanation

/var/log/syslog does not exist on the assigned VM, so the systemd journal (journalctl -p warning) was used as the equivalent source, reviewed on 02 Oct 2026 at 01:22 WAT. Warnings were low risk and operational in nature. The inetssim service relies on a deprecated SysV init script without a native systemd unit, indicating legacy configuration that should be modernised or removed if unused. Additional warnings related to realtime scheduling (rtkit-daemon) and kernel integrity-measurement (IMA) behaviour. No warnings indicated active compromise or unauthorised activity.

Event and Condition	Evidence Source	Time	User / Service	Security Significance	Recommended Action
Deprecated SysV init script for inetssim	journalctl (-p warning)	01 Oct 23:22 to 23:37	systemd-sysv-generator / inetssim	Legacy, unmaintained service increases configuration risk	Replace with a native systemd unit or remove the service if unused
Realtime scheduling thread starved	journalctl (-p warning)	01 Oct 23:36:09	rtkit-daemon	Operational/performance issue, not a direct security risk	Monitor; investigate if it recurs
IMA duplicate measurements not logged	journalctl (-p warning)	02 Oct 00:22:41	Kernel	Minor gap in integrity-measurement logging	Review IMA configuration if integrity monitoring is required
No syslog file on host	ls output	02 Oct 01:22	System Configuration	Traditional log tooling will see nothing; logs are journal-only	Forward journal to SIEM or enable file-based logging

7. Module 3 Linux Security Assessment with Lynis

Lynis is an open-source auditing tool for Unix-like systems. It performs configuration and security checks and reports warnings, suggestions and a hardening index. It is an audit and assessment tool rather than an active penetration-testing tool.

Activity 3.1 Install or Access Lynis

Use the package or installation method approved for your assigned lab VM. If Lynis is already installed, document its version and proceed. If not, obtain the current stable release using the instructor-approved repository or the official Lynis/CISOfy distribution source.

If your instructor specifically directs use of the source archive workflow, use a dedicated directory such as /opt/lynis and verify the downloaded version before extraction.

```

(icdfa@icdfa)-[~]
$ date
lynis --version
Fri 02 Oct 2026 01:41:08 WAT
3.1.4

(icdfa@icdfa)-[~]
$

```

Activity 3.2 Run a System Audit

From the installed Lynis location, run the system audit using the appropriate command for your installation. Common package installations support:

```
sudo lynis audit system
```

Source-directory installations commonly use:

```
sudo ./lynis audit system
```

Record the key results, including the hardening index (where shown), warnings, suggestions and the report/log locations.

```

File Actions Edit View Help
(icdfa@icdfa)-[~]
$ sudo lynis audit system

[ Lynis 3.1.4 ]
#####
###
Lynis comes with ABSOLUTELY NO WARRANTY. This is free software, and you are
welcome to redistribute it under the terms of the GNU General Public License.
See the LICENSE file for details about using this software.

2007-2024, CISofy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)
#####
###

[+] Initializing program
- Detecting OS ... [ DONE ]
- Checking profiles ... [ DONE ]

```

```

File Actions Edit View Help
- Report data : /var/log/lynis-report.dat

=====
Lynis 3.1.4

Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)

2007-2024, CISOfy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)

=====

[TIP]: Enhance Lynis audits by adding your settings to custom.prf (see /etc
/lynis/default.prf for all settings)

(icdfa@icdfa)-[~]
$

```

The audit is complete and the above screenshot is the Lynis version and audit execution evidence which shows Lynis 3.1.4 and the report location (/var/log/lynis-report.dat).

Activity 3.3 Optional Safe Hardening and Retest

Choose one low-risk Lynis recommendation that you understand and that is permitted in your assigned VM. Record the current state, implement the approved improvement, and re-run the relevant check or Lynis audit.

Do not make a change merely to increase a score. Explain the control objective, operational impact and how you verified the change.

Required Evidence Module 3

- Lynis version and audit execution evidence.
- Hardening index or equivalent summary, if reported.
- At least three warnings/suggestions or notable findings.
- For each selected finding: risk/control issue, evidence, likely owner, recommended remediation and priority.
- Optional retest evidence showing before/after state for one approved hardening action.

```
icdfa@icdfa: ~  
File Actions Edit View Help  
Home DEL AB  
2007-2024, CISofy - https://cisofy.com/lynis/  
Enterprise support available (compliance, plugins, interface and tools)  
  
[TIP]: Enhance Lynis audits by adding your settings to custom.prf (see /etc  
/lynis/default.prf for all settings)  
  
(icdfa@icdfa)-[~]  
$ date  
sudo grep "hardening_index" /var/log/lynis-report.dat  
sudo grep "^warning\\[\\]" /var/log/lynis-report.dat  
Fri 02 Oct 2026 03:37:09 WAT  
[sudo] password for icdfa:  
hardening_index=65  
warning[]=FIRE-4512|iptables module(s) loaded, but no rules active  
  
(icdfa@icdfa)-[~]  
$
```

```
(icdfa@icdfa)-[~]  
$ sudo grep "report_datetime_end" /var/log/lynis-report.dat  
report_datetime_end=2026-10-02 03:25:35  
  
(icdfa@icdfa)-[~]  
$
```

Summary

- Lynis version: 3.1.4
- Hardening index: 65/100, which is a moderate posture with clear room to improve
- Warnings: 1
- Report and log locations: /var/log/lynis-report.dat and /var/log/lynis.log
- The audit finished at 2026-10-02 03:25:35 WAT

ID	Findings and Risks	Evidence	Likely owner	Recommended Remediation
W4-F02	No active host firewall rules. iptables modules are loaded but no rules exist, so the host doesn't filter inbound or outbound traffic.	Lynis FIRE-4512 (the only warning)	System Administrator / Network Security	Enable a host firewall (ufw or nftables) with default-deny inbound and allow only required services
W4-F03	Weak password policy. There is no minimum or maximum password age, no password-strength module, no expiry dates and no hashing-round configuration. This raises the risk of credential compromise.	Lynis AUTH-9286, AUTH-9262, AUTH-9282, AUTH-9230	IAM / Security Administrator	Configure password aging in /etc/login.defs and install a PAM strength module (e.g. libpam-passwdqc)
W4-F04	No automated response to repeated authentication failures. This links to your Module 2 evidence of three historical authentication failures, which are only logged and never acted on.	Lynis DEB-0880 (fail2ban not installed) plus journal authentication failures	Security Operations / System Administrator	Install and configure fail2ban, or alert on a failure threshold

Lynis 3.1.4 completed a system audit on 02 Oct 2026 (finished 03:25:35 WAT) and reported a hardening index of 65/100 with one warning and multiple suggestions. The highest-priority condition is the absence of active firewall rules (FIRE-4512), which leaves the host without network traffic filtering. Further weaknesses were identified in password policy (AUTH-9286, AUTH-9262, AUTH-9282) and automated brute-force protection (DEB-0880). Together, these indicate partially implemented preventive controls, where monitoring evidence exists but enforcement and response controls require strengthening.

8. Module 4 SIEM, Automation and Continuous Monitoring

This section is conceptual. You are not required to deploy a SIEM or SOAR platform during this laboratory. The objective is to connect host-level evidence to enterprise monitoring and governance.

SIEM Capabilities

- Log collection and aggregation from multiple systems.
- Normalization and parsing of different log formats.
- Correlation and analysis of related events.
- Alerting and notification based on rules or analytics.
- Dashboards and reporting for security operations, management and compliance.

Modern monitoring ecosystems may also include UEBA, SOAR, cloud integrations, EDR, network traffic analysis, vulnerability scanners and threat-intelligence platforms.

Automation in Monitoring and Auditing

- Continuous Control Monitoring: automated or near-real-time verification that selected controls remain effective.
- Automated Alerting: notification when security events or control thresholds are breached.
- Trend Analysis: systematic identification of changes over time.
- Automated Reporting: recurring generation of control/compliance reports.
- Automated Control Testing: scripts or integrations that test configurations or evidence conditions.
- Workflow Automation: assignment, tracking, escalation and follow-up of findings or remediation actions.

SIEM / Continuous Monitoring Mapping

Linux Evidence	How it reaches the SIEM	SIEM Use	Technical Alert (SOC)	Governance and Control Issue
auditd events	audit dispatcher plugin or agent (e.g. Wazuh/Splunk forwarder)	Correlate user, command, file	Write to shadow/passwd; auditd stopped	Audit coverage gaps; rules not loaded
journalctl (auth, sudo, errors)	Journal export or rsyslog/agent	Failed-login trends; privileged-use reports	>5 failures in 10 min; unusual sudo user	Auth source missing from monitoring
Lynis report	Parse /var/log/lynis-report.dat on a schedule	Dashboard: hardening index, warnings over time	n/a (periodic, not real-time)	Index below target; overdue findings
Firewall/service state	Scheduled script or agent check	Continuous control testing	Firewall disabled unexpectedly	Control absent on multiple hosts

Governance Discussion

Explain how the evidence you collected from auditd, journalctl, authentication logs and Lynis could be forwarded to or represented in an enterprise monitoring environment. Identify what should remain a technical operational alert and what conditions should become a governance/control-assurance issue.

Host-level security information can be sent to a SIEM using agents, audit tools, or journal exports. The SIEM can then organise and compare the data from different systems. Urgent events, such as repeated failed login attempts, changes to account files, or audit services being stopped, should be treated as technical alerts for the SOC to investigate and respond to quickly.

However, issues that continue over time and show that a security control is missing or not working properly should be treated as governance concerns. Examples include having no active firewall, missing

authentication logs, a low hardening score, or overdue security fixes. These issues should be assigned to an owner, given a clear threshold and review date, and monitored through management dashboards and regular control-assurance reports.

9. Applied Governance Task From Linux Evidence to Control Assurance

Using the evidence collected in Modules 1–3, identify at least three events, weaknesses or conditions that warrant governance attention. These may include repeated authentication failures, unexpected privileged activity, missing audit coverage, overdue hardening issues, inadequate logging, critical Lynix findings or another evidence-based condition.

Control-Monitoring Table

Control / Objective	Evidence	Owner	Observed Status	KPI/KRI or Threshold	Risk / Significance	Remediation	Retest / Follow-up
Audit logging of sensitive files and privileged use (accountability)	auditctl -l, ausearch and aureport: passwd_changes, shadow_changes, program_executions firing; audit=icdfa preserved under sudo	Security Operations / SysAdmin	Effective	100% of required rules loaded; any write to passwd/shadow outside a change window triggers an alert	Actions are attributable to a named user. Read-level watches are noisy (617 passwd events)	Keep rules; tune read events before SIEM forwarding	Monthly auditctl -l check; after reboot
Authentication log coverage	/var/log/auth.log missing; auth_failures rule watches a non-existent file	SysAdmin / SecOps	Gap: false assurance	100% of auth sources monitored	Empty results would look like "no failures" when there is no coverage	Replace with journal-based monitoring or enable a file-based auth log	Generate a benign failed login and confirm it is captured
Failed-authentication monitoring and response	Journal: 3 failures for icdfa (10 May, 20 Sep, 24 Sep); Lynix DEB-0880 (no fail2ban)	Security Operations	Partially effective: logged, not acted on	Alert: >5 failures per account in 10 min; escalate if privileged/root account	Isolated, likely mistyped, but no automated response exists	Install and configure fail2ban or a SIEM alert rule	Simulated failures trigger an alert or ban
Host firewall (network protection)	Lynix warning FIRE-4512: iptables loaded, no rules active	SysAdmin / Network Security	Not effective	0 hosts without an active default-deny policy	Unfiltered inbound and outbound traffic	Enable ufw or nftables, default-deny, allow required services only	Re-run Lynix; FIRE-4512 should clear
Password policy (credential strength)	Lynix AUTH-9286, AUTH-9262, AUTH-9282, AUTH-9230	IAM / Security Admin	Partially implemented	100% of accounts with max age per policy (e.g. 90 days) and strength enforcement	Weak or non-expiring passwords raise compromise risk	Set aging in /etc/login.defs; install libpam-passwdqc	Re-run Lynix; verify with chage -l

The table above contains five monitored control/evidence rows with evidence collected from this laboratory.

Governance Escalation Questions

1. Which finding is the highest priority and why?

Highest priority: The missing firewall (FIRE-4512) should be addressed first because it is the only Lynis warning and means an important preventive security control is completely absent. The authentication log gap should be the next priority because it can affect the organisation's ability to detect suspicious activity.

2. Who should own remediation for each major finding?

Control owners: The firewall should be handled by the SysAdmin/Network Security team, while password policies should be managed by IAM. Authentication monitoring and Fail2Ban should be handled by Security Operations, with audit coverage managed jointly by SecOps and SysAdmin.

3. What threshold or condition should trigger escalation to the CISO, Risk function or management committee?

Escalation triggers: A High-severity finding should be escalated to the CISO if it remains unresolved for more than 30 days. Other situations that should trigger escalation include unexpected changes to `/etc/shadow` or `/etc/passwd`, `auditd` stopping unexpectedly, or more than five failed login attempts on a privileged account within 10 minutes. The issue should be escalated to the Risk function or management committee if a critical control gap affects multiple systems, a remediation deadline is missed twice, or the hardening index falls below the agreed target, such as 70.

4. What evidence would demonstrate successful remediation?

Evidence of successful remediation: The fix should be supported by clear evidence. This could include a new Lynis scan showing that FIRE-4512 has been resolved and the hardening score has improved, `ufw` status showing that the firewall is active, and `chage -l` confirming that the new password-aging settings are in place. A simulated failed login should also generate an alert or trigger a ban, while the journal-based authentication logs should be confirmed as visible in the monitoring system.

5. When and how should the control be retested?

Each fix should be tested immediately after implementation by rerunning the relevant security check or a Lynis scan. The controls should then be reviewed regularly, with monthly checks for areas such as `auditd` rules and firewall status and a quarterly full Lynis scan. All test results and remediation evidence should be recorded in the control register.